Sebastian De Leon Lab 6 - Prelab

Task 1 assuming shift by 1 bit

1. 0x FF FF FF FF

shift left 0xF = 1111 0x FF FFFF FF N=1 Z=0 C=1 V=0

shift right 0xFF FFFFFE

shift right 0 → 0x 7F FF FF FF 1 N=0 V=0 Z=0 C=1

rotate right 1 0x FF FF FF FF N=1 C=1 Z=0 V=0

rotate left 1 0xFF FF FF FF N=1 C=1 V=0 Z=0

2. 0x C5 C5 C5 C5

1100 0101

1001 1010

shift left N=0 V=0 C=1 Z=0

shift right C 5 C 5

1100 0101

0 → 0110 0010 1

N=0 C=1 V=1 Z=0

rotate right 1100 0100 → 1

1110 0010

N=1, V=0

Z=0, C=1

rotate left

1 ← 11000101 10000101

1000 1011

N=1 Z=0

V=0 C=1

0x A3 A3 A3 A3 A 3 3

1010 0011 0011 ← 0 N = 0

V = 0

C = 1

Z = 0

shift left

A 1010 [1] ← 0100 0110

S = 0011

shift right 1010 10100011

→ 0 0101 0001 → [1] N = 0

V = 0

C = 1

Z = 0

rotate right

[1] →

1010 0011 N = 1

V = 0

1101 0001 C = 1

Z = 0

rotate left

[1] →

1010 0011 ← N = 0

V = 0

0100 0111 C = 1

Z = 0

0x 2F 36 2F 36 2

0010 1111 0110 0010

shift left 2 F 3 6 2

← 0 N = 0

V = 0

0101 0100 C = 0

Z = 0

shift right 0010 1111 0110 0010

→ 0 0001 0001 N = 0 Z = 0

V = 0

0110 C = 0

rotate right → 0010 0010

[0] ←

0001 0001 N = 0 V = 0 C = 0 Z 0

rotate left

[0] →

0010 0010

0101 0100 N = 0 V = 0

C = 0 Z = 0

3E7D3E7

1101 0111

shift left

1101 0011 0111 ← 0

1 ← 1010 1110

N = 1
V = 0
C = 1
Z = 0

1110

shift right

1101 0011 0111 → 1

0 → 0110 0011

N = 0
V = 0
C = 1
Z = 0

rotate right

1101 0011 1110 0111

1110 0011

N = 1 C = 1
V = 0 Z = 0

rotate left

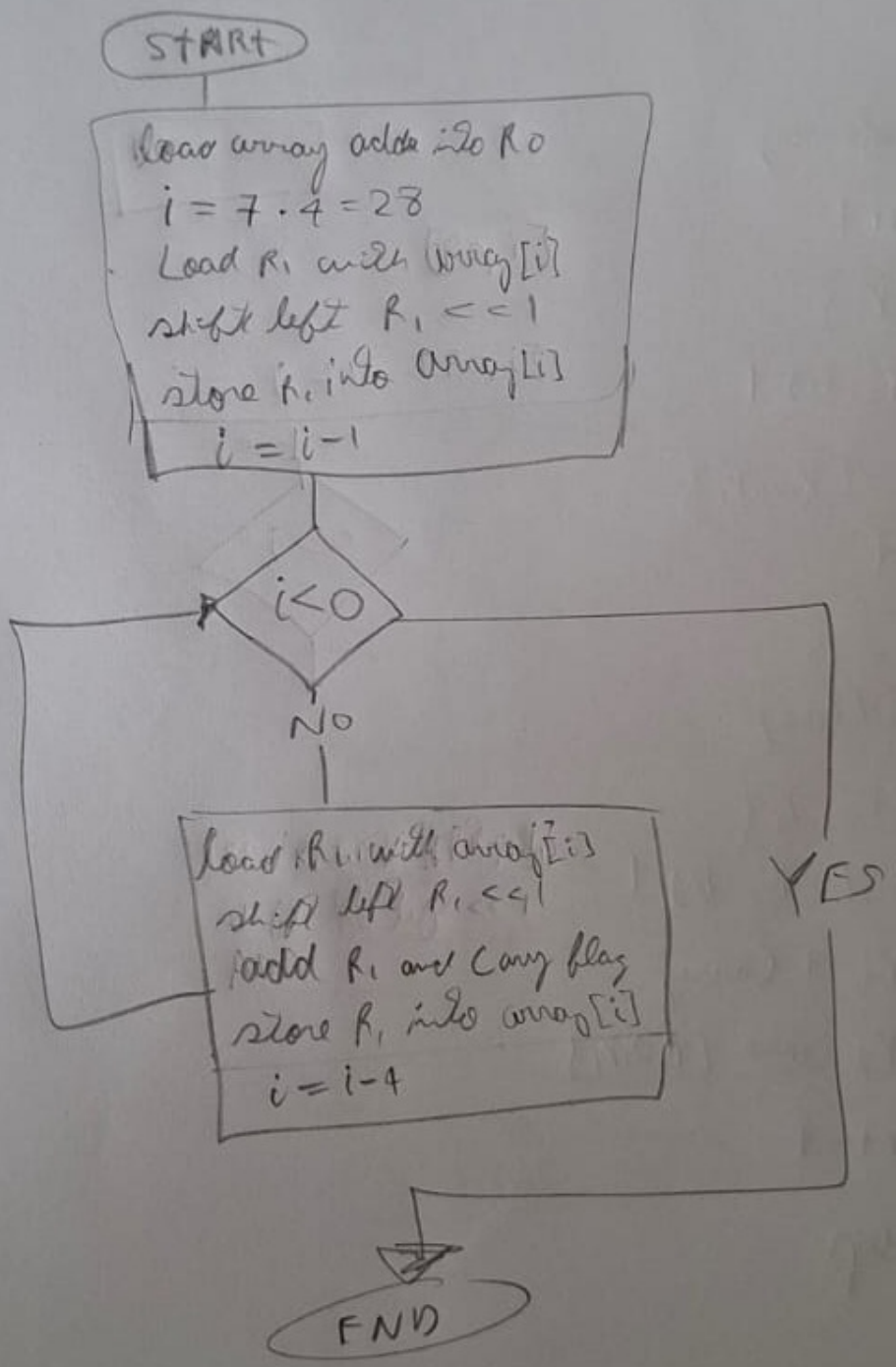
1101 0011 1110 0111

1010 1 1111

N = 1 C = 1
V = 0 Z = 0

1	2	3	4	5
FC 3F	00 00 00 A2	00 00 FF FE	00 00 C5 23	00 00 36 AD

6	7	8
00 00 00 0F	04 1D 42 35	2F CC 2F CC



DATA

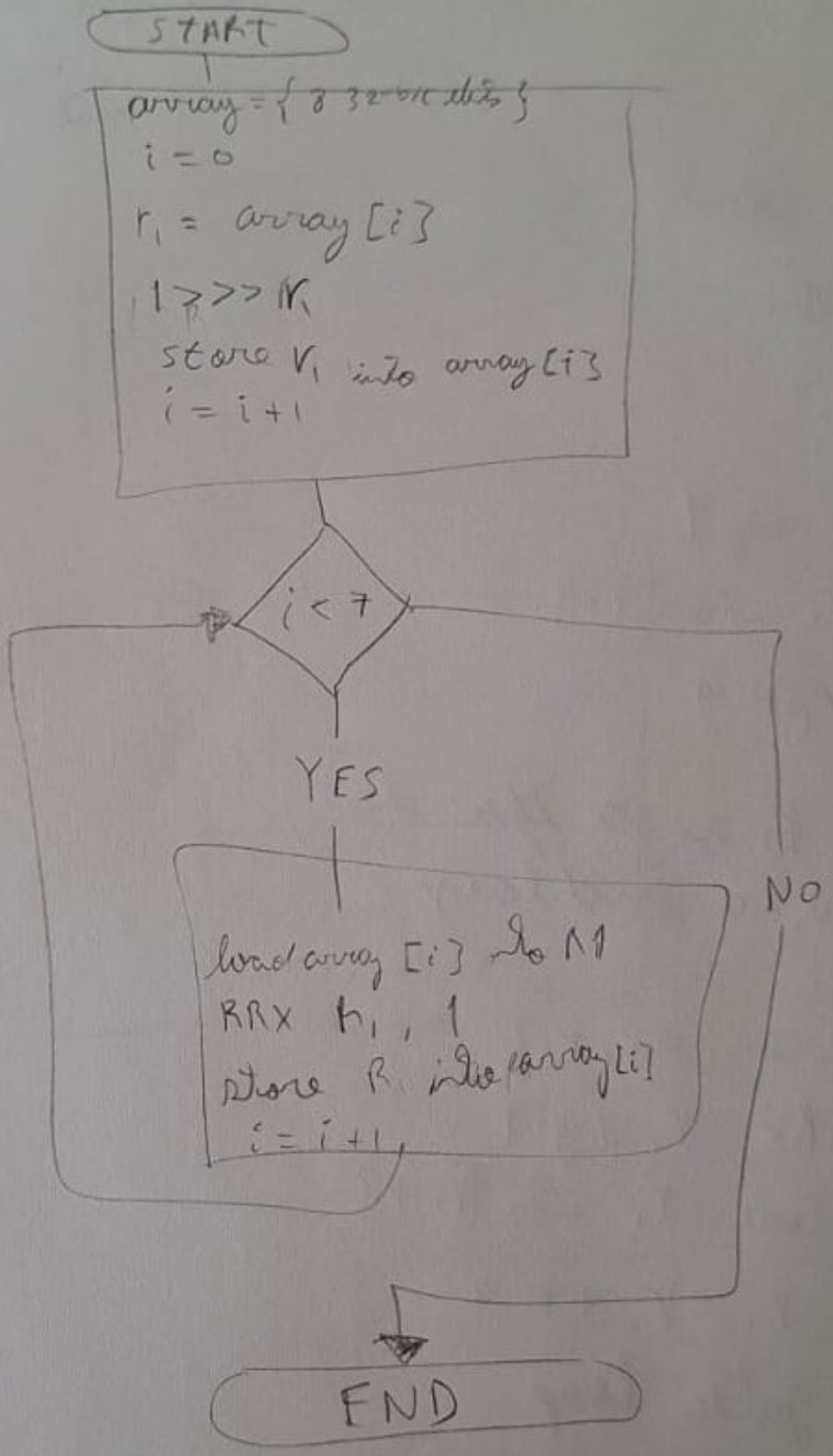
array = {...}

Counter = 28

CODE

```
r0 = address of array
r1 = Counter
r2 = [r0, r1]
r2 = LSL r2 by 1
store R2 [r0, r1]
r1 = r1 - 4
loop  if r1 < 0
      B endloop
      r2 = [r0, r1]
      r2 = LSL R2 by 1
      r2 = r2 + Carry
      STR r2 into [R0, r1]
      r1 = r1 - 4
      B loop
endloop end
```

shifting to the right



DATA

array = {...}

counter = 0

CODE

r_0 = addr of array

r_1 = counter

r_2 = [r_0]

ASR r_2 by 1

store r_2 into [r_0]

r_1 = r_1 + #4

loop

if r_1 >= 32 then do

goto endloop

else

r_1 = [r_0, r_1]

RRX r_1 by 1

store r_1 into [r_0, r_1]

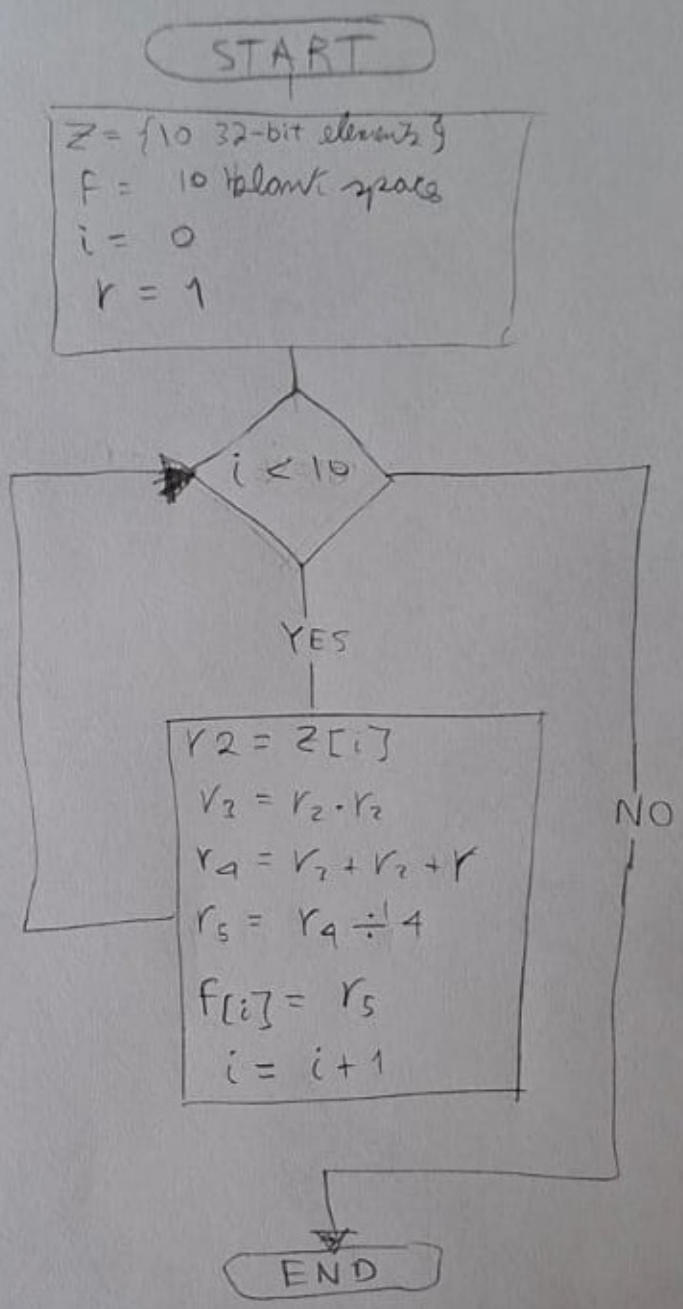
r_1 = r_1 + #4

goto loop

endloop

Farray in which each elent is equal to

$0.25 \cdot (Z[i]^2 + Z[i] + r)$ $\quad r = 1$



Pre lab task 3 code

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab6\prelab\prelab.uvprojx - µVision [Non-Commercial Use License]

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Target_1

Project

Project: prelab
Target_1
Source Group 1
CMSIS

main.s

```
;ARM assembly code shell

        PRESERVE8
        THUMB

; Vector Table
        AREA    RESET, DATA, READONLY
        EXPORT  __Vectors

__Vectors
        DCD     0x20018000          ; Top of Stack (96KB RAM)
        DCD     Reset_Handler       ; Reset Handler

        AREA    arrayData, DATA, READONLY
source_array    DCD     0x1, 0x2, 0x3, 0x4, 0x5, 0x6, 0x7, 0x8, 0x9, 0xA; declarin

        AREA    myData, DATA, READWRITE
dest_array      space   40
f_array         space   40
size            equ     40  ; declaring size of array
index           equ     0   ; declaring index
constant        equ     1

        AREA    |.text|, CODE, READONLY

; Reset Handler
Reset_Handler   PROC
                EXPORT  Reset_Handler
                BL      main
                B       .
                ENDP


main            PROC
                EXPORT  main

                ldr r0, =source_array   ; loading array address in r0
                ldr r1, =dest_array
                ldr r10, =f_array
                mov r2, #index          ; current index

copy_arr_loop
                cmp r2, #size
                BGE end_copy_arr_loop
                ldr r3, [r0, r2]
```

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab6\prelab\prelab.uvprojx - µVision [Non-Commercial Use License]

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Target_1

oject

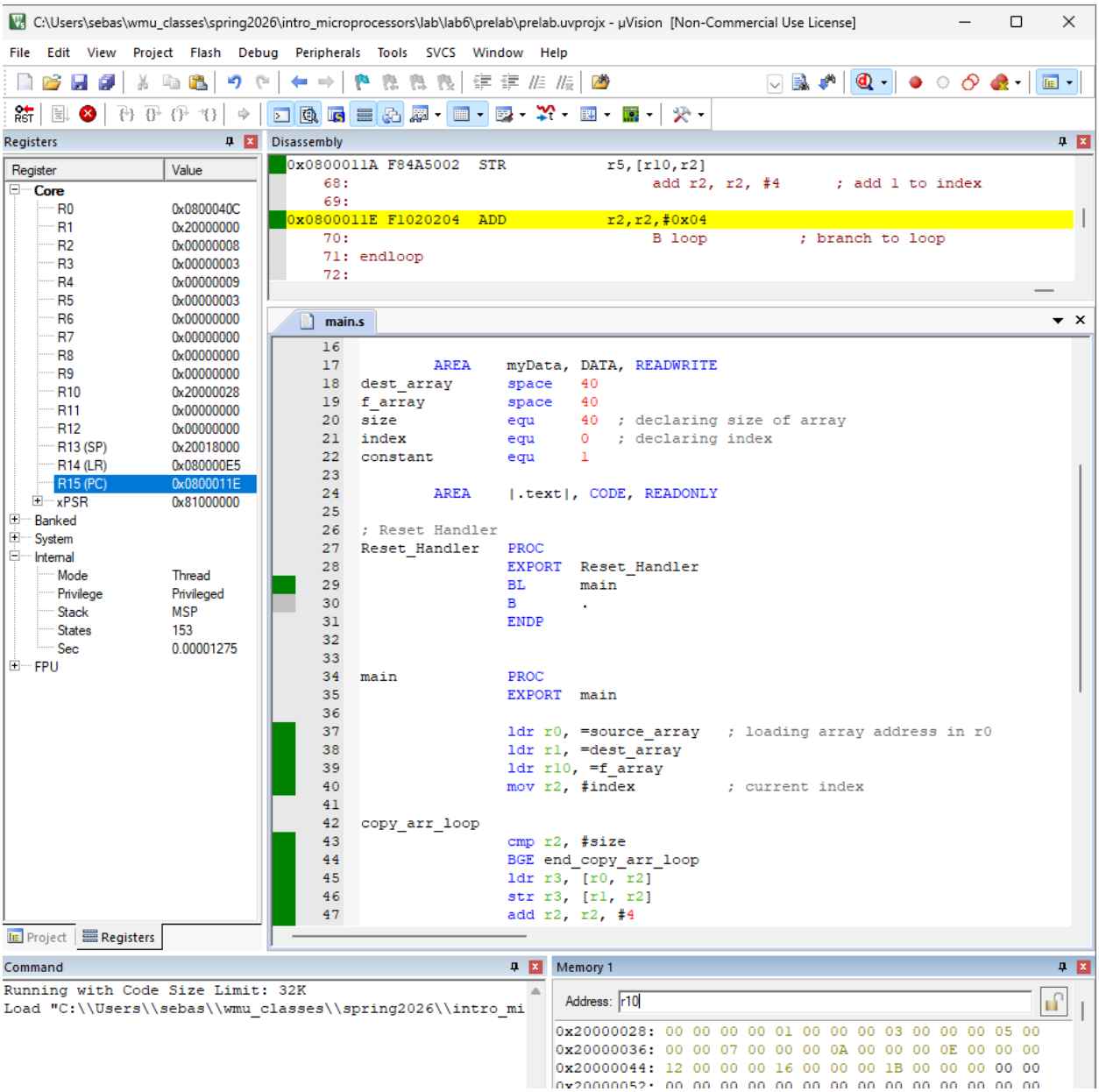
Project: prelab
- Target_1
  - Source Group 1
  - CMSIS

main.s

```
main            PROC
                EXPORT  main

                ldr r0, =source_array   ; loading array address in r0
                ldr r1, =dest_array
                ldr r10, =f_array
                mov r2, #index          ; current index

copy_arr_loop
                cmp r2, #size
                BGE end_copy_arr_loop
                ldr r3, [r0, r2]
                str r3, [r1, r2]
                add r2, r2, #4
                B copy_arr_loop
end_copy_arr_loop

                mov r2, #index          ; reseting r2 index to 0

loop            cmp r2, #size           ; comparing index to the size

                BGE endloop             ; if index >= size end loop

                ; element operation

                ldr     r3, [r1, r2]
                mul     r4, r3, r3
                add     r5, r4, r3
                add     r5, r5, #constant
                lsr     r5, #2
                str     r5, [r10, r2]

                ; end element operation

                add r2, r2, #4          ; add 1 to index

                B loop                  ; branch to loop
endloop

                BX      LR              ; Return from main
                ENDP

                ALIGN
                END
```

uild Output

```
uild started: Project: prelab
** Using Compiler 'V6.24', folder: 'C:\Users\sebas\AppData\Local\Keil_v5\ARM\ARMCLANG\Bin'
uild target 'Target_1'
ssembling main.s...
inking...
rogram Size: Code=996 RO-data=80 RW-data=80 ZI-data=96
.\Objects\prelab.axf" - 0 Error(s), 0 Warning(s).
uild Time Elapsed:  00:00:00
```

Simulation

Prelab task 3 result